5b. Kepler Length

- **kepler_length.cpp**: compute the distance travelled by a planet along an elliptical orbit by evaluating the integral:

$$L = \int_0^t v(\tau)\, d\tau$$

- Here the speed v(t) has the following expression

$$v(t) = \alpha a \sqrt{\frac{1 + e\cos(E(t))}{1 - e\cos(E(t))}}$$

where $\alpha = \sqrt{GM/a^3}$, $0 < e < 1$ is the eccentricity (also: a is the semi-major axis, M is the stellar mass, and the planet is at pericentre at t = 0), while E(t) is the *eccentric anomaly* given by the unique root of the following equation:

$$f(E,t) = E - e\sin(E) - \alpha t = 0$$

- Set $\alpha = a = 1$ and choose the eccentricity `e = 0.85`. Use both the 3 point Gauss-Legendre *and* the trapezoidal quadrature rules to evaluate the integral for $0 \le t \le 10$. Repeat each integration using n=10, n=100 and n=1000 equal sub-intervals in $t \in [0,10]$ and print the resulting values of L using scientific notation with 12 digits after the decimal point.

- Choose the secant method to solve for the eccentric anomaly E(t) with a tolerance of 10^{-10}.

- Use macros to define eccentricity and α (e.g., `#define ECCENTRICITY 0.85`) and declare a global variable to define time (e.g. `static double g_time`).

- Upload your **.cpp** file with i) the output inserted in the comments at the beginning of the file, ii) the required library functions at the end, e.g.

```
// Name: <last name>, <first name>
// Date: <date >
// Code output:
// ***************************************************
// Gauss L(n=10)   = ...
// Trapz L(n=10)   = ...
// Gauss L(n=100)  = ...
// Trapz L(n=100)  = ...
// Gauss L(n=1000) = ...
// Trapz L(n=1000) = ...
// ***************************************************
#include ...
...
int main()
{
  // code here
}

double GaussianQuad(...){}
double TrapezoidalQuad(...){}
double Secant(...){}
```